

Data and Analytics

Use Case 6: Real-Time Data Processing

Scenario: Analyze streaming data from IoT devices.

Tasks:

- o Use Kinesis Data Streams to ingest data.
- o Use Lambda or Glue for data processing.
- o Store processed data in S3 and visualize it using QuickSight.

Step 1: Setup Amazon Kinesis Data Streams for Ingestion

Amazon Kinesis Data Streams captures streaming data from IoT devices in real time.

1. **Create a Kinesis Data Stream:**
 - o Go to the **Kinesis Console**.
 - o Click **Create data stream**.
 - **Stream name:** IoTDataStream.

- **Number of shards:** Start with 1 (can scale based on throughput requirements).
- Click **Create data stream**.

Step 2: Process Data Using AWS Lambda or AWS Glue

1. Create a Lambda Function:

- Go to the **Lambda Console** and click **Create function**.
- Choose **Author from scratch**.
 - **Function name:** IoTDataProcessor.
 - **Runtime:** Python, Node.js, or your preferred language.
 - **Role:** Create a new role with **AWSLambdaBasicExecutionRole** and Kinesis permissions.

2. Configure Kinesis Trigger:

- Under **Function Overview**, click **Add trigger**.

- Select **Kinesis** and configure:
 - **Stream:** IoTDataStream.
 - **Batch size:** Default (e.g., 100).
 - **Starting position:** Trim Horizon (to start from the oldest records).

Step 3: Store Processed Data in Amazon S3

1. Create an S3 Bucket:

- Go to the **S3 Console** and click **Create bucket**.
 - **Bucket name:** iot-processed-data.
 - **Region:** Same as Kinesis.

2. Enable Encryption:

- Turn on **S3 Bucket Encryption** using SSE-S3 or SSE-KMS.

3. Apply Bucket Policies:

- Restrict access to specific IAM roles and IP ranges.
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Step 4: Visualize Data Using Amazon QuickSight

1. Setup QuickSight:

- Go to the **QuickSight Console** and sign up.
- Configure **S3 as a data source**:
 - Select `iot-processed-data` bucket.
 - Specify the folder path if necessary.

2. Prepare Data Set:

- Use QuickSight's data preparation features to clean and organize the data.
- Aggregate data by metrics such as time, device ID, or temperature.

3. Create Visualizations:

- Build dashboards with charts (e.g., line charts for temperature trends, bar charts for device counts).

- Share dashboards with stakeholders.

Step 5: Security Optimizations

1. IAM Roles:

- Use specific roles with minimal permissions for Lambda, Glue, and Kinesis.
- Example:
 - Lambda Role: Allow Kinesis GetRecords, GetShardIterator, and S3 PutObject.

2. Data Encryption:

- Enable encryption for:
 - Kinesis: Use KMS for encrypting stream data.
 - S3: Enable SSE-S3 or SSE-KMS.

3. Network Security:

- Use VPC endpoints for Kinesis and S3 to avoid exposing data to the public internet.
 - 4. **Logging and Monitoring:**
 - Enable **CloudWatch Logs** for Lambda and Kinesis.
 - Use **CloudTrail** to monitor API activity.
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Step 6: Cost Optimization

1. **Right-Size Resources:**
 - Scale Kinesis shards based on actual throughput.
 - Use Lambda reserved concurrency to control costs.
2. **Data Storage:**
 - Configure **S3 Lifecycle Rules** to transition processed data to S3 Glacier for long-term storage.
3. **QuickSight Pricing:**

- Use the **Standard Edition** for basic analytics.
 - Delete unused datasets and dashboards.
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Step 7: Test and Validate

1. **Simulate IoT Data:**

- Generate test data from IoT devices to verify the entire pipeline.

2. **Monitor Metrics:**

- Use CloudWatch to monitor Kinesis throughput, Lambda invocations, and S3 storage.

3. **Dashboard Validation:**

- Verify the accuracy of data visualizations in QuickSight.